Technical Report Writing on

Error Detection and Correction at the Data Link Layer

Submitted by

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1. Abstract

Error detection and correction at the Data Link Layer ensure reliable data transmission between nodes. Techniques like parity check, checksum, and CRC detect errors, while Hamming code and ARQ correct them. These methods add redundancy to identify corrupted bits and maintain data integrity during communication over noisy channels.

2. Introduction

The Data Link Layer is the second layer of the OSI model and plays a crucial role in ensuring reliable communication between directly connected devices. When data is transmitted over a physical medium, it is vulnerable to various types of errors caused by noise, interference, attenuation, and signal distortion. These disturbances may alter one or more bits in a frame, leading to incorrect or corrupted data at the receiver's end. To maintain data integrity and reliability, the Data Link Layer implements error detection and correction mechanisms.

Error detection techniques such as parity check, checksum, and Cyclic Redundancy Check (CRC) are used to identify whether a frame has been corrupted during transmission. Among these, CRC is widely used due to its high accuracy in detecting burst errors. In addition to detecting errors, the layer may also employ error correction methods like Hamming code and Automatic Repeat reQuest (ARQ), which either correct the errors directly or request retransmission of damaged frames. These mechanisms collectively ensure efficient and dependable data communication across networks.

3. Procedure and Discussion

Error Detection and Correction at the Data Link Layer:

The Data Link Layer (Layer 2 of the OSI model) is responsible for reliable node-to-node communication. One of its main tasks is detecting and correcting errors that occur during data transmission over the physical medium.

- **Why Errors Occur?**

Errors occur during data transmission due to noise, electromagnetic interference, signal attenuation, distortion, and hardware imperfections. External factors like weather, cross-talk between cables, or faulty connectors can also alter transmitted bits. These disturbances change signal characteristics, causing the receiver to interpret incorrect binary values, resulting in corrupted data frames.

- **Error Control in Data Link Layer:**

Error control in the Data Link Layer ensures reliable and accurate delivery of frames between directly connected nodes. It detects and corrects errors that occur during transmission over the physical medium.

The Data Link Layer (Layer 2 of the OSI model) uses two main approaches for error control:

1) Error Detection

To detect errors, a common technique is to introduce redundancy bits that provide additional information. Various techniques for error detection include:

- Simple Parity Check
- Checksum
- Cyclic Redundancy Check (CRC)

1. Simple Parity Check

Simple Parity Check is the most basic error detection technique used in data communication at the Data Link Layer (Layer 2 of the OSI model).

It works by adding one extra bit, called the parity bit, to the original data.

Types of Parity :

i. Even Parity

The total number of 1s (including the parity bit) must be even.

Example:

Data = 1011 → (three 1s = odd)

Parity bit = 1
Transmitted data = 10111 (now four 1s = even)

ii. Odd Parity

The total number of 1s must be odd.

Example:

Data = 1010 → (two 1s = even)

Parity bit = 1

Transmitted data = 10101 (three 1s = odd)

2. Checksum

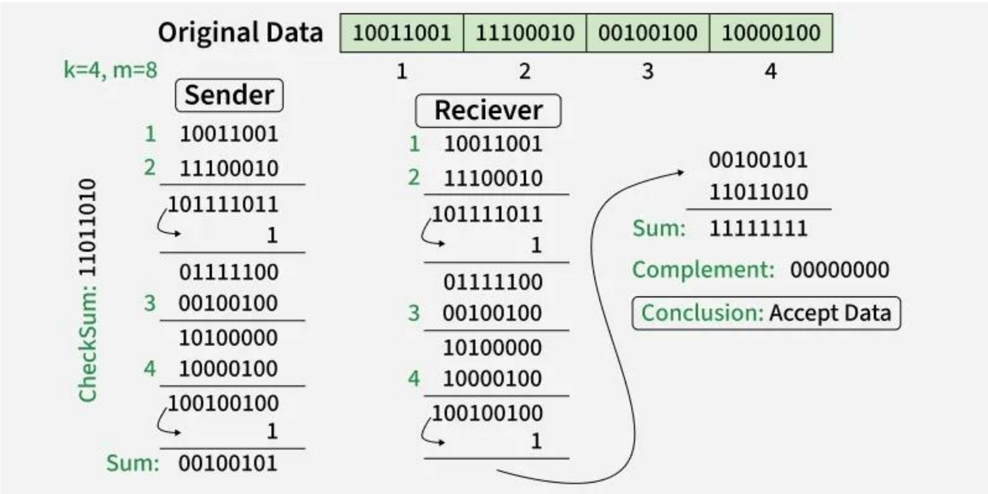
Checksum detects errors by adding data segments using 1's complement arithmetic. The result is sent with the data. The receiver repeats the calculation to verify integrity and detect any transmission errors.

At Sender:

1. Divide data into equal parts.
2. Add all parts using 1's complement addition.
3. Take 1's complement of the result.
4. Attach it to the data and transmit.

At Receiver:

1. Add all received segments including checksum.
2. If result is all 1s → No error.
3. If result is not all 1s → Error detected.



3. Cyclic Redundancy Check (CRC)

Cyclic Redundancy Check (CRC) is a powerful error detection technique used in digital networks and storage devices. It is widely implemented at the Data Link Layer (Layer 2 of the OSI model).

<u>Working Principle:</u>

We have given dataword of length n and divisor of length k.

Step 1: Append (k-1) zero's to the original message

Step 2: Perform modulo 2 division

Step 3: Remainder of division = CRC

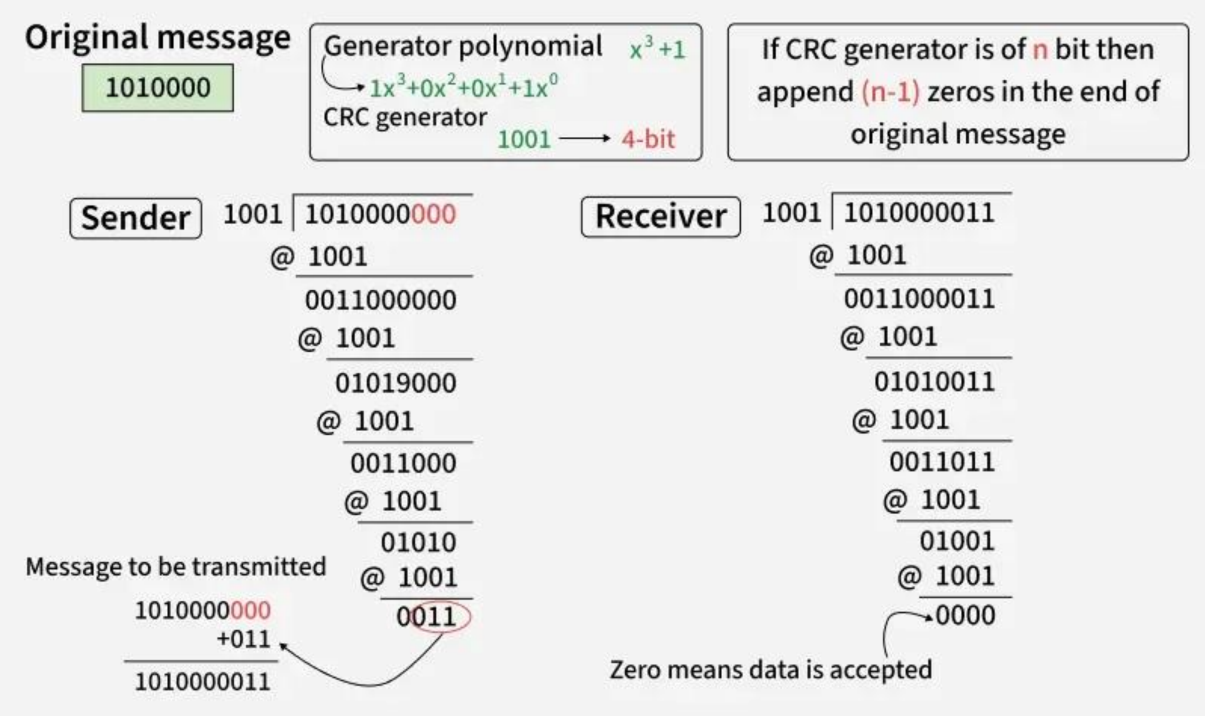
Step 4: Code word = Data with append k-1 zero's + CRC

Note:

CRC must be k-1 bits

Length of Code word = n+k-1 bits

Example: Let's data to be send is 1010000 and divisor in the form of polynomial is x3+1. CRC method discussed below.



2) Error Correction

Error correction is a technique used in computer networks to detect and correct errors that occur during data transmission. It ensures reliable communication over noisy channels. Error correction is mainly handled at the Data Link Layer (Layer 2 of the OSI model).

Types of Error Correction

1. Forward Error Correction (FEC):

- The sender adds extra redundant bits.
- The receiver detects and corrects errors without retransmission.
- Used in wireless, satellite, and real-time communication.
- Example: Hamming Code

i) Hamming Code

Hamming Code is an error correction technique used to detect and correct single-bit errors in data transmission. It is a type of Forward Error Correction (FEC) method and was developed by Richard Hamming.

Formula:

To calculate the number of parity bits (r):

$$\mathbf{2^r \geq m + r + 1}$$

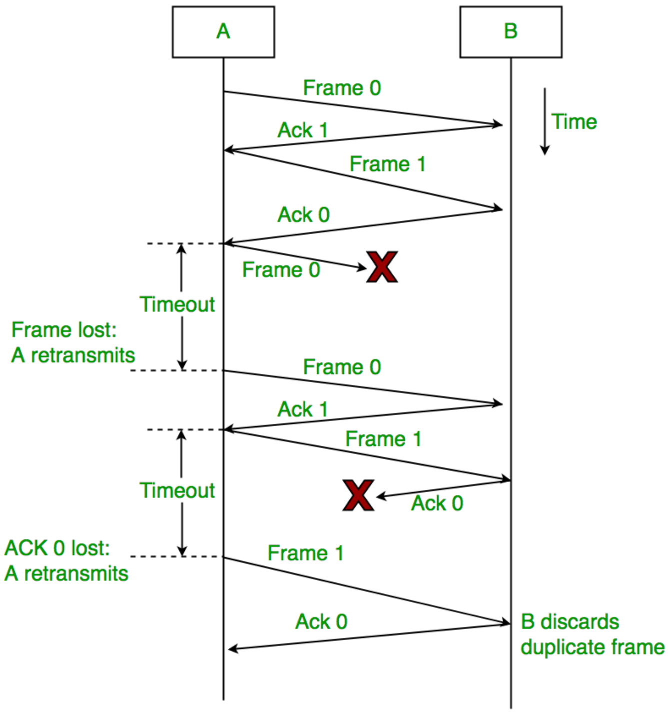
- Hamming Code can correct one-bit errors and detect two-bit errors, but it cannot fix two errors at the same time.
- It is suitable for systems where errors are rare and usually occur one bit at a time.2. Reed–Solomon Code
- Parity bits are placed at positions 1, 2, 4, 8, 16, …, which are powers of two.
- Each parity bit checks a specific group of data bits, helping locate the exact position of an error.
- The receiver uses parity checks (syndrome) to identify the incorrect bit and correct it if needed.
- Since the receiver can fix the error by itself, no retransmission is required, making it a true Forward Error Correction technique.

2. Automatic Repeat reQuest (ARQ):

Automatic Repeat reQuest (ARQ) is an error control technique used in computer networks to ensure reliable data transmission. It combines error detection with retransmission. If the receiver detects an error in a frame, it requests the sender to retransmit the corrupted data.

i) Stop-and-Wait ARQ:

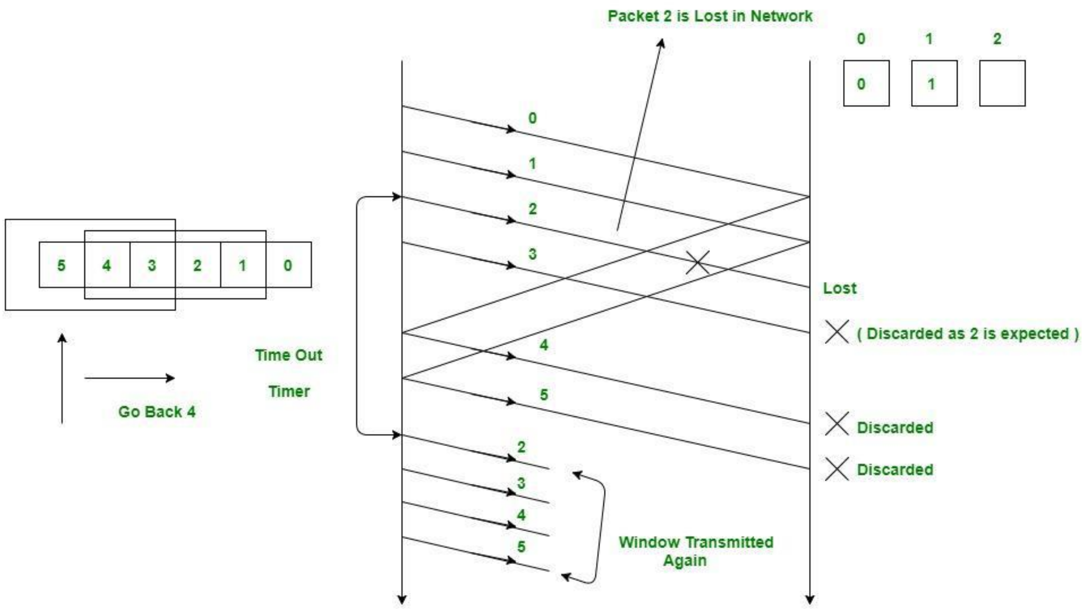
In Stop-and-Wait ARQ, the sender transmits one data frame at a time and waits for an ACK (Acknowledgment) from the receiver. If the ACK is not received within a time limit, the sender retransmits the frame. Once the ACK is received, the sender sends the next frame. This process continues until all frames are successfully delivered.



ii) **Go-Back-N ARQ:**

It is an error control protocol in which the sender is allowed to send multiple frames (up to a fixed window size) without waiting for ACKs.

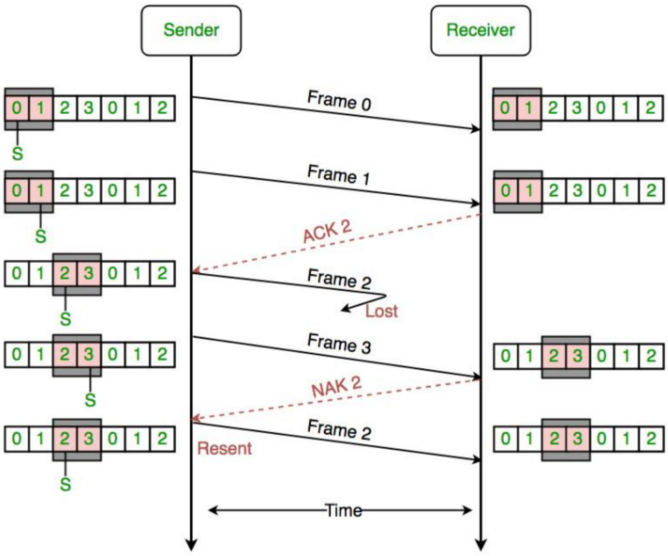
- The receiver sends ACKs for correctly received frames.
- If a frame is lost or arrives with an error, the receiver discards that frame and all subsequent frames.
- The sender then retransmits the erroneous frame along with all the frames sent after it.
- When no errors occur, the protocol behaves like a normal sliding window mechanism.



Go-Back-N ARQ

iii) Selective Repeat ARQ:

It is an error control protocol in which only the frames that are lost or corrupted are retransmitted, unlike Go-Back-N where multiple frames are resent.



- It improves efficiency by avoiding unnecessary retransmissions.
- The sender retransmits only those frames for which a NAK is received or a timeout occurs.
- Each frame is acknowledged individually, allowing correctly received frames to be accepted and stored.
- Due to the need for additional buffering and tracking, the protocol is more complex, which limits its use compared to simpler ARQ schemes.

➢ Advantages of Error Detection and Correction at the Data Link Layer :

1. Improves Data Integrity

Detects and corrects corrupted bits, ensuring accurate data delivery.

2. Increases Reliability

Techniques like CRC and ARQ ensure frames are correctly received.

3. Reduces Data Loss

Retransmission of corrupted frames prevents permanent data loss.

4. Efficient Communication

Prevents unnecessary higher-layer processing of corrupted data.

5. Supports Noisy Channels

Works effectively even in environments with interference and signal distortion.

4. Conclusion

Error detection and correction at the Data Link Layer play a vital role in ensuring reliable communication between network devices. By using techniques like CRC, Hamming Code, and ARQ, the layer detects corrupted frames and corrects or retransmits them. These mechanisms improve data integrity, reduce transmission errors, and maintain efficient and stable network communication.

5. References

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